

轮趣科技

Self-balancing unmanned bicycle based on handlebar steering control

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1. Kinetic model

As an unmanned two-wheeled intelligent transportation tool, the unmanned bicycle has the advantages of flexible movement, energy efficiency and The high utilization rate and green environmental protection make it unique in the fields of security patrol and avoiding traffic congestion.

However, on the other hand, it is a typical representative of static instability and dynamic stability.

In short, a bicycle cannot maintain a stable balance without any control.

Therefore, based on the dynamic model of the bicycle, this paper designs a controller to make the bicycle maintain

A state of balance.

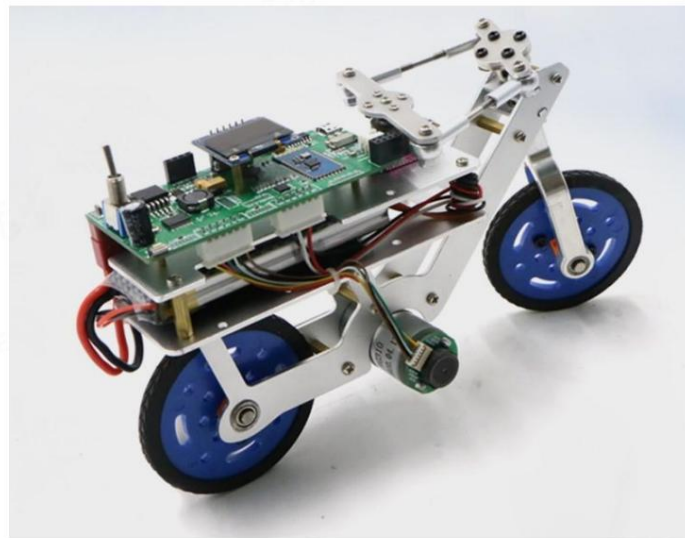


Figure 1-1 Physical picture of a self-balancing unmanned bicycle

For ease of analysis, let's simplify the bicycle model. Assume that the bicycle moves on a horizontal plane.

Assume that the front fork axis is perpendicular to the ground. The simplified model is shown in Figure 1-2.

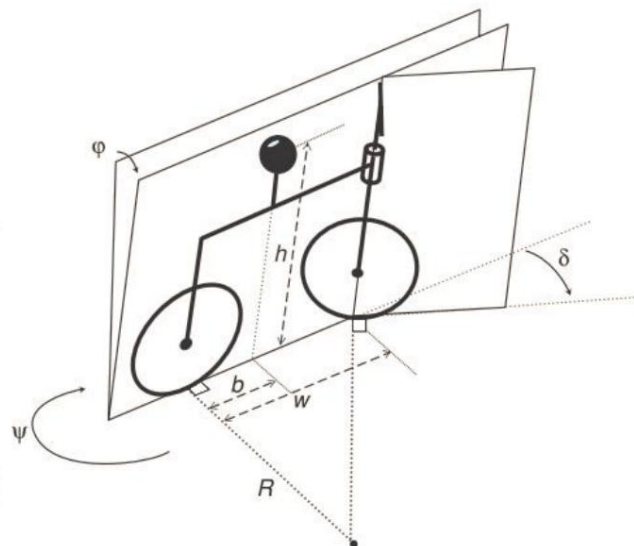


Figure 1-2 Bicycle model [1]

Assume that the mass of the vehicle is m , the height of the center of mass is $\bar{y}$, the inclination angle of the vehicle is ϕ , the front fork angle is δ , and the front and rear wheels are

The distance between the contact point with the ground is R , where the distance between the landing point of the rear wheel and the projection of the center of gravity on the ground is b .

The distance is w . Establish the following moving coordinate system $O\bar{y}XYZ$ as shown in the figure below.

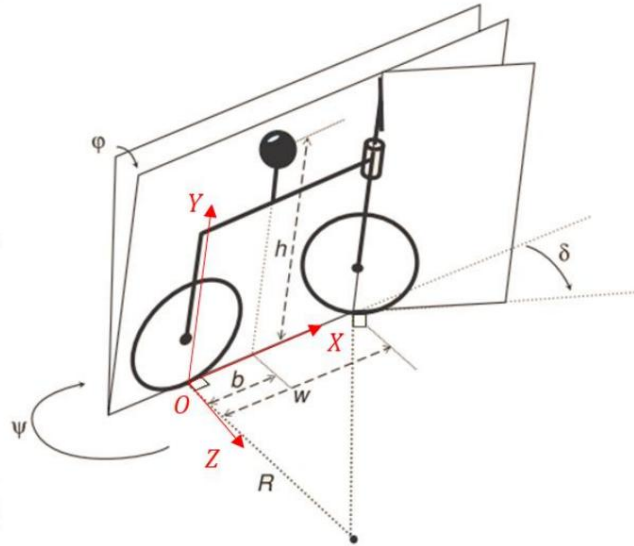


Figure 1-3 Establishment of the moving coordinate system

Since the front fork turning angle is relatively small, it can be considered that the turning radius of the center of mass movement is approximately equal to the distance from the rear wheel to the center of the circle.

The distance between O and C is w . The following relationship is satisfied:

$$\tan \phi = \frac{h}{w} \quad (1)$$

Right now

$$\frac{h}{w} = \tan \phi \quad (2)$$

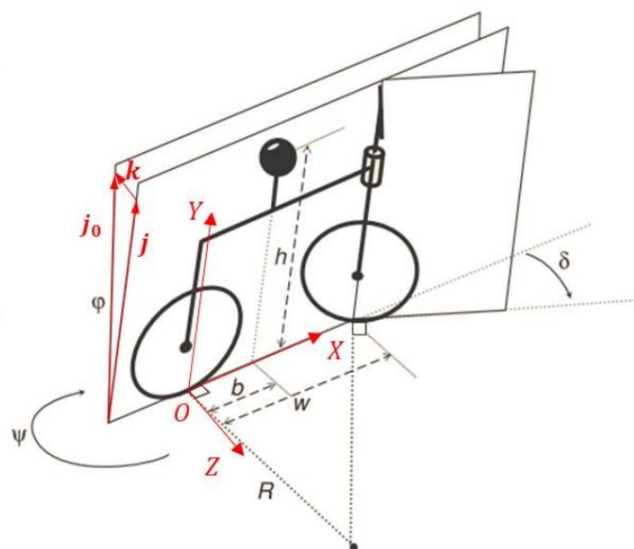


Figure 1-4 Relationship between the basis vectors of the vertical axis and the basis vectors of each axis

Assume that the basis vectors of each axis of the coordinate system (vectors are represented by blue bold fonts here) are $\bar{i}, \bar{j}, \bar{k}$, The centroid

The position (relative to the origin O of the follower coordinate system, i.e. the landing point of the rear wheel) can be expressed as $x = r \sin \gamma$.

The angular velocity of the vehicle body is composed of the angular velocity of rotation around the horizontal axis (X axis) and the angular velocity of rotation around the vertical axis (not the Y axis).

The angular velocity consists of two parts, among which the angular velocity of rotation around the horizontal axis is

The velocity is. (Figure 1-3 shows how the basis vector of the vertical axis is represented by the basis vectors of each axis.

Here is a simple explanation, $\dot{x} = r \cos \gamma \dot{\gamma}$, and because the inclination angle of the car body is relatively small,

$\dot{x} \approx r \dot{\gamma}$. In fact, analogous to circular motion, $\dot{x}$ is obtained by the rotation angle, $\dot{\gamma}$ is the position

Obviously, when $\dot{\gamma} \neq 0$, the direction of displacement (or the direction of velocity) is the tangent of the point.

The direction of the line, that is, $\dot{\gamma}$, is close to the arc length, that is. Therefore, $\dot{x} \approx r \dot{\gamma}$, that is, $\dot{x} = r \dot{\gamma}$

So, there is

$$\dot{x} = r \dot{\gamma} \quad (3)$$

in

$$\dot{x} = r \dot{\gamma} \quad (4)$$

Then the velocity of the center of mass relative to the origin O of the moving coordinate system is

$$\dot{x} = r \dot{\gamma} + \dot{y} \quad (5)$$

The actual velocity of the center of mass can be obtained as

$$\dot{x} = r \dot{\gamma} + \dot{y} \quad (6)$$

From equation (2) and equation (4), we know that

$$\dot{x} = r \dot{\gamma} \quad (7)$$

And because $\dot{\gamma} \neq 0$, so $\dot{\gamma} \neq 0$.

Therefore (only first-order small quantities are considered in the calculation)

$$\dot{x} = r \dot{\gamma} \quad (8)$$

Considering that the bicycle moves at a constant speed on a horizontal plane, we have the following constraints:

$$\begin{cases} \dot{\gamma} = 0 \\ \dot{\gamma} = 0 \end{cases} \quad (9)$$

From equations (8) and (9), we can get the acceleration of the bicycle (the basis vector in the moving coordinate system is not a constant vector)

$$\dot{x} = r \dot{\gamma} \quad (10)$$

Assuming that the mass of the bicycle is concentrated at the center of mass, the weight of the bicycle body and the centrifugal force can be expressed as

Theorem of moment of momentum about the X axis under the action of inertial force[2]

$$[\times ()] \ddot{y} = 0 \quad (11)$$

Combining equations 7 and 11, we can get the dynamic equation of the vehicle body rotating around the horizontal axis:

$$= \frac{2}{\ddot{y}} \quad (12)$$

So the state equation of the system is

$$\begin{pmatrix} \dot{y} \\ \ddot{y} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{2}{\ddot{y}} & 0 \end{pmatrix} \begin{pmatrix} y \\ \dot{y} \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{2}{\ddot{y}} \end{pmatrix} \quad (13)$$

On the other hand, the output equation of the system is

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y \\ \dot{y} \end{pmatrix} \quad (14)$$

Equations 13 and 14 are the state space expressions of the system.

2. Design the controller

2.1 LQR controller

LQR (Linear Quadratic Regulator) is a linear quadratic regulator. LQR can get the state line

The optimal control law of linear feedback is easy to form closed-loop optimal control. LQR optimal control takes advantage of low cost

It can make the original system achieve better performance indicators and is easy to implement. It is a relatively mature development in modern control theory.

The following is a brief overview of its principle and then applies it to the balancing process of the unmanned bicycle.

middle.

Assume that the state space expression of the continuous linear time-invariant system (assuming the system is fully controllable) is

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases} \quad (15)$$

In modern control theory, the most basic control system is the full-state feedback control system (the system block diagram is shown in the figure

2-1), whose goal is to design a state feedback controller $\hat{y}$ to control the performance of the system.

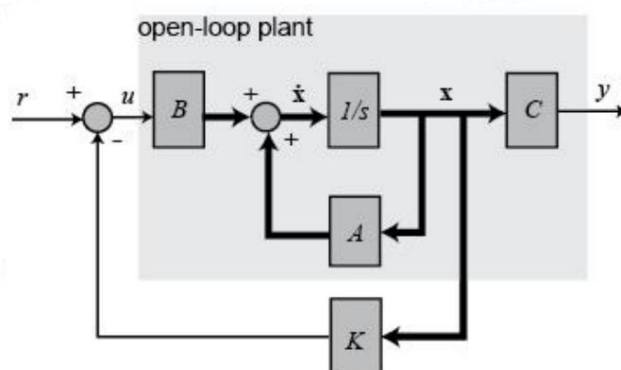


Figure 2-1 Full state feedback control system

The problem now is that this is not the only one, so what is the best one?

Before we look at the cost function, let's look at this problem (from the MATLAB official tutorial): Suppose you are looking for

The best way to get from home to work, and there are many ways to choose from, you can drive, ride a bicycle

So, which mode of transportation is the best choice for you?

There is no answer in itself, because we don't know what is "best". Assume that we only consider time cost and money cost.

Money cost, fill in the information of 4 modes of transportation into the following table.

Table 2-1

Transportation	Time / min	Money / \$
Car	20	45
Bike	75	0

Bus	30	12
Helicopter	4	2578

As we can see, if time cost is our biggest concern, then renting a helicopter is obviously the best choice.

On the other hand, if we are more concerned about the monetary cost, then riding a bicycle would be the best choice.

In life, people tend to seek a balance between the two. For example, if you have a morning meeting today, you will naturally

Pay attention to the time cost, but on the other hand, you are not rich enough, so you should also pay attention to the money cost.

So how do you choose? Here is a simple quantitative evaluation method to create such an evaluation form.

Number = $\ddot{y}$ + $\ddot{y}$, set different weights according to personal needs. If the weight is set as

As shown in Table 2-2, then the best choice is to drive to work. If the weights are set as shown in Table 2-3, then,

The best option is to go to work by bus.

Table 2-2

Transportation Q	Time/min R	10	20	1	10	1	Money / \$	J(Cost)
Car							45	245
Bike				75			0	750
Bus	10		30		1		12	312
Helicopter	10		4		1		2578	2618

Table 2-3

Transportation Q	Time / min R			Money / \$	J(Cost)
Car	5	20	5	45	325
Bike	5	75	5	0	375
Bus	5	30	5	12	210
Helicopter	5	4	5	2578	12910

Similarly, for control systems, we introduce the cost function (Quadratic Cost Function)

$$= \int_0^{+\infty} [\ddot{y}^T () () + \ddot{y}^T () ()] \quad (16)$$

In the formula, is the semi-positive definite state weight matrix, is the positive definite control weight matrix.

In , and are symmetric matrices and are usually diagonal matrices. The elements on the diagonal represent the corresponding error components

(The ultimate goal is to make the value of each state variable become 0, so the value of each state variable

The more important the error component is, the faster it is expected to decrease.

The larger the weight coefficient is, the greater the weight coefficient is. Similarly, the elements on the diagonal in represent the corresponding input components

The design idea of LQR controller is to design a state feedback controller

= $\ddot{y}$, so that the cost function is minimized, thereby achieving the goal of using cheap costs to make the original system achieve better results

The purpose of performance indicators. As for how to calculate the state feedback matrix, here are only the corresponding conclusions:

$$= \dot{y}_1^T \quad (17)$$

Where, is the **Riccati** equation $\dot{P} + P\dot{y} + \dot{y}_1^T P = 0$ solution.

Similarly, for a discrete linear time-invariant system (assuming the system is fully controllable), let its state equation be

$$(x_{k+1} = Ax_k + Bu_k) \quad (18)$$

The quadratic performance index is

$$J = \sum_{k=0}^{N-1} [x_k^T Q x_k + u_k^T R u_k] \quad (19)$$

Wherein, Q is a symmetric positive definite constant matrix or a symmetric semi-positive definite constant matrix, and R is a symmetric positive definite constant matrix.

The problem now is to design a state feedback controller $u_k = -Kx_k$, which minimizes. As above,

Here we only give the corresponding conclusions:

$$K = (P^{-1}B^T + \dot{y}_1^T) \quad (20)$$

where is the **Riccati** equation $\dot{P} + P\dot{y} + \dot{y}_1^T P + (P^{-1}B^T + \dot{y}_1^T)B = 0$ solution.

With the above qualitative description of the LQR controller, let's take the balancing process of a bicycle as an example.

Explain the design of the LQR controller. From the dynamic model of the bicycle, we can see that the state equation of the system is

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -\frac{g}{h} & 0 & \frac{v}{h} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \frac{1}{h} \\ 0 \end{pmatrix} \quad (13)$$

The output equation is

$$y = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \quad (14)$$

Normally, we do not calculate the state feedback matrix manually, but use the relevant functions in MATLAB.

The relevant procedures are as follows:

```

clc
g = 9.8; % Gravity acceleration
h = 0.088; % Height of the bicycle's center of mass
v = 0.634; % Bicycle speed
w = 0.167; % Distance between the front and rear wheels and the ground contact point
b = 0.055; % The distance between the point where the rear wheel touches the ground and the projection of the center of gravity on the ground
A_21 = g/h;
A_23 = -v^2/(w*h);
B_21 = -(b*v)/(w*h);
A = [0 1 0; A_21 0 A_23; 0 0 0];
B = [0; B_21; 1];
C = [1 0 0; 0 0 1];
  
```

```

D = 0;
Ts = 0.020; t = % Sampling interval
0:Ts:4; u =
zeros(size(t)); [G,H] =
c2d(A,B,Ts); x0 = [0.0873; % Convert the continuous system to a discrete
0; 0]; Tc = ctrb(G,H); if system % Set the initial state of the system
(rank(Tc)==3)
fprintf('This system is
controllable\n'); Q = [300 0 0; 0 0 0; 0 0
300]; % Q matrix R = 1; % R matrix K = dlqr(G,H,Q,R);
% Calculate the state feedback coefficient (check the
calculated value yourself) G2 = GH*K; = dlsim(G2,H,C,D,u,x0);

y
subplot(2,1,1)
plot(t,y(:,1),'b-.','LineWidth',1.5); xlabel('Time(s)');
ylabel('\phi(rad) ');

grid on
subplot(2,1,2)
plot(t,y(:,2),'b-.','LineWidth',1.5); xlabel('Time(s)');
ylabel('\delta( rad)');

grid on
end

```

The results after running are as follows:

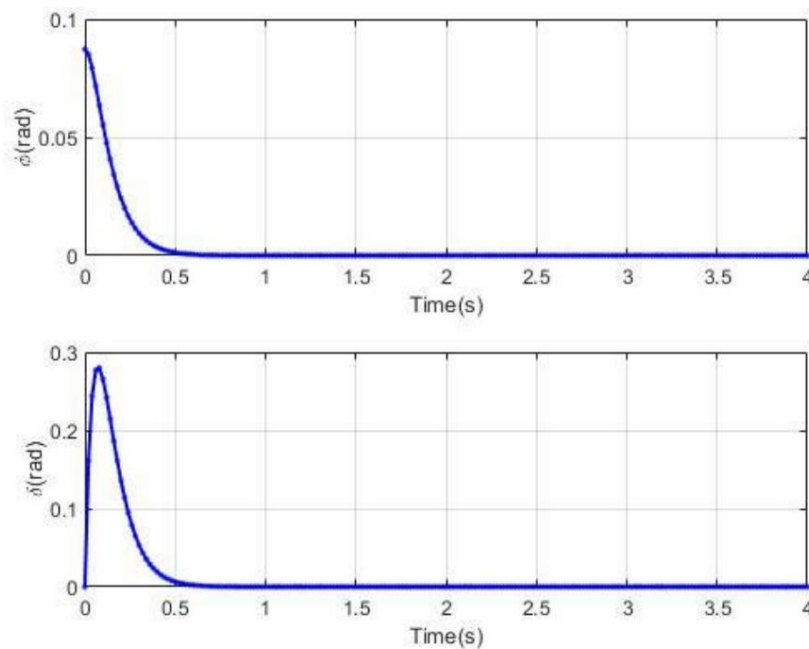


Figure 2-2 Discrete-time system LQR simulation results

The corresponding C language code is as follows:

```
float k1=-92.2973, k2=-8.6746, k3=10.5355; //LQR feedback coefficient
...
delta_dot = -( k1*(phi-balance_point) + k2*phi_dot + k3*delta );
...
```

2.2 PID Controller

Since the output of the LQR controller, the steering speed of the fork, can be directly controlled by the rudder

The servo is obtained by the user (it is impossible to effectively model the servo) and the servo is an open-loop system for the user.

Therefore, it is considered to add a PID controller at the output of the LQR controller.

In our problem, the control rate can be easily obtained:

$$PWM = \begin{bmatrix} 1 \\ 0 \end{bmatrix} p' \begin{bmatrix} \delta \\ \dot{\delta} \end{bmatrix} + d' \left(\begin{bmatrix} \delta \\ \dot{\delta} \end{bmatrix} \ddot{\delta} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \ddot{\delta} \right) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \ddot{\delta} \quad (equation 21)$$

Among them, PWM corresponds to the steering angle of the servo, indicating the deviation, which is determined by the steering speed of the front fork.

$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ to act as a brake, because when the bicycle is tilted more, the output of the LQR controller will be

Therefore, the output of the LQR controller is used as the input of the PID controller in order to make the LQR

The output of the controller changes to 0 as quickly as possible, and the corresponding system deviation is also 0. Therefore, equation (21) can be changed to

$$PWM = \begin{bmatrix} 1 \\ 0 \end{bmatrix} p' \begin{bmatrix} \delta \\ \dot{\delta} \end{bmatrix} + d' \left(\begin{bmatrix} \delta \\ \dot{\delta} \end{bmatrix} \ddot{\delta} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \ddot{\delta} \right) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \ddot{\delta} \quad (equation 22)$$

Further

$$PWM \ddot{\delta} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} p' \begin{bmatrix} \ddot{\delta} \\ \dot{\delta} \end{bmatrix} + d' \left(\begin{bmatrix} \ddot{\delta} \\ \dot{\delta} \end{bmatrix} \ddot{\delta} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \ddot{\delta} \right) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \ddot{\delta} \quad (equation 23)$$

Subtracting Equation 22 from Equation 23 yields the expression for incremental PID control:

$$PWM = PWM \ddot{\delta} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} p' \left(\begin{bmatrix} \ddot{\delta} \\ \dot{\delta} \end{bmatrix} \ddot{\delta} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \ddot{\delta} \right) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \ddot{\delta} \quad (equation 24)$$

The corresponding C language code is as follows:

```
int ki=4, kp=16, kd=4; //PID parameters (after magnification 100 times, integers are easy to debug)
...
u = ki*delta_dot + kp*(delta_dot-last_delta_dot) +
kd*(delta_dot-2*last_delta_dot+last_delta_dot_2);
Servo += u/10000;
...
```

3. References

- [1] Limebeer D, Sharp R S. Bicycles, motorcycles, and models[J]. IEEE control systems, 2006, 26(5):34-61.
- [2] Liu Yanzhu. On the stability of bicycles[J]. Mechanics and Practice, 2012, 34(2):4.

4. Appendix

4.1 About Steering Structure

Regarding the steering structure of a bicycle, we idealize it into a trapezoidal structure, as shown in the following figure:

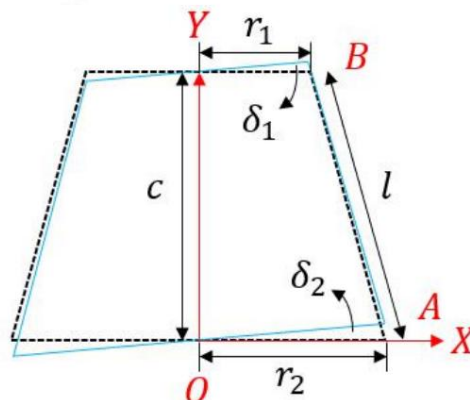


Figure 4-1 Steering structure of a self-balancing bicycle

When stationary, there is $\vec{r}_1 = (r_1, 0)$, $\vec{r}_2 = (r_2, 0)$; after rotation, we have $\vec{r}_1 = (r_1 \cos \delta_1, r_1 \sin \delta_1)$ and $\vec{r}_2 = (r_2 \cos \delta_2, r_2 \sin \delta_2)$. Since the distance between A and B remains unchanged during the rotation, we have

$$l^2 = \left((r_1 \cos \delta_1 - r_2 \cos \delta_2)^2 + (r_1 \sin \delta_1 - r_2 \sin \delta_2)^2 \right) \quad (25)$$

Expand

$$l^2 = r_1^2 (1 - \cos \delta_1 \cos \delta_2) + r_2^2 (1 - \cos \delta_1 \cos \delta_2) + 2 r_1 r_2 \sin \delta_1 \sin \delta_2 = 0 \quad (26)$$

When the rotation angle of the steering gear and the bicycle handlebar is small, it can be considered $\delta_1 \approx 0$, $\delta_2 \approx 0$. Then, that (equation) 26 can be simplified to

$$\delta_1 = \frac{r_2}{r_1} \delta_2 \quad (27)$$

Therefore, it can be approximately assumed that the bicycle handlebar angle is proportional to the steering gear angle.

4.2 About the system block diagram

The system block diagram of the above control system can be drawn as follows:

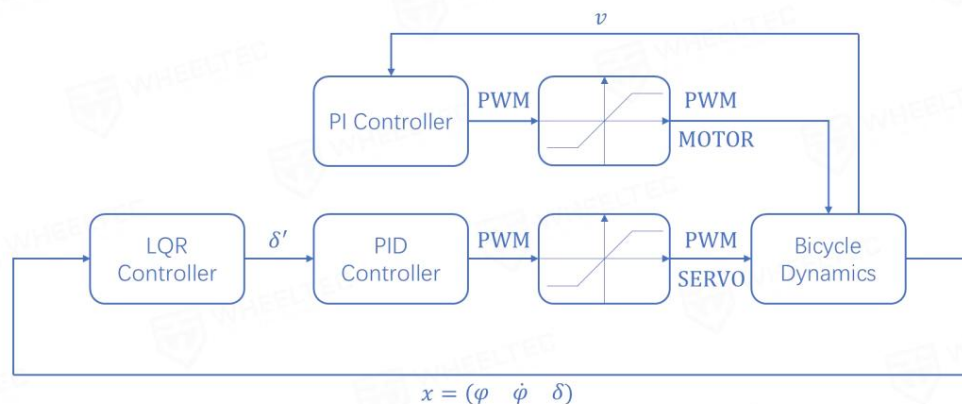


Figure 4-2 Control system block diagram